

HE2003 Econometrics I

Tutorial 10

04/11/2024

Question 1

Explain the following terms; Use specific example, formula, or graph if necessary

- a) Explain the similarity and dissimilarity between t-test and homoskedasticity-only F -test.
- b) Explain the overall regression F -test in the homoskedastic case.
- c) Given an empirical example of linear regression model with two dummy variables and their interaction term (the model also includes a constant term). Explain the meaning of all the parameters in the model.

Question 1 (a)

The formula for the t -test (e.g., for testing $H_0: \beta_1 = 0$) is:

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}$$

And the formula for the homoskedasticity-only F -test is:

$$F = \frac{(R_u^2 - R_r^2)/q}{(1 - R_u^2)/(n - k - 1)}$$

where q is the number of restrictions. When $q = 1$, then $F = t^2$

Question 1 (a)

Similarity: In both equations, the denominators play the role of standardization, and the numerator compares certain differences: in the t -test, the numerator compares the difference between the estimator $\hat{\beta}_1$ and the hypothesized values 0, and the null hypothesis is rejected when such difference is sufficiently large; in the F -test, the numerator compares the difference between the fittings of the unrestricted and restricted models, and the null hypothesis is rejected when such difference is sufficiently large (that is, the restricted model (H_0) is rejected when the fitting of the unrestricted model is much better than that of the restricted model).

Dissimilarity: There are many, for example, the t -test can only test for single restriction ($q = 1$), while the F -test can test multiple restrictions; the t -test can take negative values, while the F -test can only take non-negative values; in large samples the t -test follows $N(0,1)$ under the null hypothesis, while in large samples the F -test follows $F_{q,\infty}$ (or χ_q^2/q) under the null hypothesis, etc.

Question 1 (b)

For the linear regression model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki} + u_i, \quad i = 1, \dots, n$$

often we want to test the joint hypothesis that all the slope coefficients of the regressors $X_{1i}, \dots, X_{ki}$ are zero. That is, the null and alternative hypotheses are

$$H_0: \beta_1 = 0, \beta_2 = 0, \dots, \beta_k = 0 \text{ vs } H_1: \beta_j \neq 0 \text{ for at least one } j = 1, \dots, k$$

In this case, **the restricted regression model only has an intercept** (since all the slope coefficients are zero), and the corresponding R^2 will be zero. Therefore, we have $R_r^2 = 0$. In addition, here we have $q = k$, the homoskedasticity-only F -statistic thus becomes:

$$F = \frac{R_u^2/k}{(1 - R_u^2)/(n - k - 1)}$$

where R_u^2 is calculated from the unrestricted regression model. This is called the “overall” regression F -statistic and it follows $\chi_k^2/k (F_{k,\infty})$ when H_0 is true.

Question 1 (c)

Consider the following example:

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{1i} \times D_{2i} + u_i, \quad i = 1, \dots, n$$

where Y_i denotes individual i 's wage. D_{1i} and D_{2i} are dummy variables. $D_{1i} = 1$ if i has a bachelor's degree and $D_{1i} = 0$ otherwise. $D_{2i} = 1$ if i is female and $D_{2i} = 0$ otherwise. $D_{1i} \times D_{2i}$ is the interaction between the two dummy variables.

Then, for the population regression lines of the model, we can consider the following 4 scenarios with their corresponding parameters (coefficients):

$$(1) \{D_{1i} = 0, D_{2i} = 0\} = \{no\ bachelor, male\} : \beta_0$$

$$(2) \{D_{1i} = 1, D_{2i} = 0\} = \{bachelor, male\} : \beta_0 + \beta_1$$

$$(3) \{D_{1i} = 0, D_{2i} = 1\} = \{no\ bachelor, female\} : \beta_0 + \beta_2$$

$$(4) \{D_{1i} = 1, D_{2i} = 1\} = \{bachelor, female\} : \beta_0 + \beta_1 + \beta_2 + \beta_3$$

Then, β_1 is identified by (2)-(1), and β_2 is identified by (3)-(1).

Question 1 (c)

Finally, β_3 can be identified by using the difference-in-differences (DID) as follows:

$$\begin{aligned} & [(4) - (3)] - [(2) - (1)] \\ &= [(\beta_0 + \beta_1 + \beta_2 + \beta_3) - (\beta_0 + \beta_2)] - [(\beta_0 + \beta_1) - \beta_0] = (\beta_1 + \beta_3) - \beta_1 = \beta_3 \\ &= [\{bachelor, female\} - \{no bachelor, female\}] - [\{bachelor, male\} - \{no bachelor, male\}] \end{aligned}$$

where $[\{bachelor, female\} - \{no bachelor, female\}]$ fixes the gender as female and obtain the change in wage when one has a bachelor's degree (compared with those without a bachelor's degree) and $[\{bachelor, male\} - \{no bachelor, male\}]$ fixes the gender as male and obtain the change in wage when one has a bachelor's degree.

Therefore, β_3 corresponds to the **extra effect (interaction effect)** of both having the bachelor's degree and being female. Intuitively, if β_3 is positive, then the employers may have a favoritism for female university graduates over male university graduates (thus an extra effect in wage, which is captured by β_3), and vice versa.

Question 2

The following regression studies the tradeoff between time spent sleeping and working after controlling for education and age:

$$\widehat{sleep} = 3638.25 - 0.148 totalwork - 11.13 educ + 2.20 age, n = 706, R^2 = 0.113 \quad (1)$$

(112.28) (0.017) (5.88) (1.45)

where *sleep* and *totalwork* are measured in minutes per week, while *educ* and *age* are measured in years. Dropping *educ* and *age* from the equation gives

$$\widehat{sleep} = 3586.38 - 0.151 totalwork, n = 706, R^2 = 0.103 \quad (2)$$

(38.91) (0.017)

Suppose we want to test the joint null hypothesis that the coefficients of *educ* and *age* are zero. Compute the value of the homoskedasticity-only *F*-test using the R^2 from the two regressions. To what kind of distribution and what critical value should we compare the value of the *F*-test? Can we reject the null hypothesis at the 5% level? Explain.

Question 2

First, we collect the necessary information: $R_u^2 = 0.113$, $R_r^2 = 0.103$, $q = 2$, $k = 3$, and $n = 706$. In addition, **when the null hypothesis is true**, the homoskedasticity-only F -statistic follows the distribution $F_{2,\infty}$ (or $\chi_2^2/2$) in large samples because $q = 2$.

$$F = \frac{(R_u^2 - R_r^2)/q}{(1 - R_u^2)/(n - k - 1)} = \frac{(0.113 - 0.103)/2}{(1 - 0.113)/(706 - 4)} = 3.9572 > 3$$

Here, 3 is the 5% critical value of $F_{2,\infty}$ (“ ∞ ” means that we are using large-sample approximation to the distribution of the F -test, just same as we use large-sample approximation to the distribution of the t -test). Therefore, we can reject the joint null hypothesis at the 5% level because the value of the computed F -statistic is in the rejection region.

Question 3

This question refers to the table below, computed using data on employees in a developing country. The dataset consists of information over 10,973 full-time, full-year workers. The highest educational achievement for each worker is a high school diploma. The workers' ages range from 25 to 40 years. The dataset also contains information on the region of the country where the person lives, gender, and age.

Let AWE = logarithm of average weekly wage (in the country's currency), $High\ School$ = binary variable (1 if high school, 0 if less), $Male$ = binary variable (1 if male, 0 if female), Age = (in years), $North$ = binary variable (1 if Region = North, 0 otherwise), $East$ = binary variable (1 if Region = East, 0 otherwise), $South$ = binary variable (1 if Region = South, 0 otherwise), and $West$ = binary variable (1 if Region = West, 0 otherwise).

Question 3

- a) Using the regression results in column (1):
 - i. Is the high school earning difference estimated from this regression statistically significant at the 5% level? Explain. Construct a 95% confidence interval of the difference.
 - ii. Is the male-female earning difference estimated from this regression statistically significant at the 5% level? Explain. Construct a 95% confidence interval of the difference.

- b) Using the regression results in column (2):
 - i. Is age an important determinant of earnings? Use an appropriate statistical test and confidence interval to explain your answer.
 - ii. Suppose Alvo is a 30-year-old male high-school graduate, and Kal is a 40-year-old male high-school graduate. Construct a 95% confidence interval for the predicted difference between their earnings.

Question 3

Hint for (b): Hint: Consider changing X_1 by a given amount, ΔX_1 . The expected change in Y associated with this change in X_1 is $\beta_1 \Delta X_1$. Then, given the estimate $\hat{\beta}_1$, the 95% confidence interval for the effect of changing X_1 by ΔX_1 can be expressed as:

$$\left[\left(\hat{\beta}_1 - 1,96 \times SE(\hat{\beta}_1) \right) \Delta X_1, \left(\hat{\beta}_1 + 1,96 \times SE(\hat{\beta}_1) \right) \Delta X_1 \right]$$

- c) Using the regression results in column (3): Are there any important regional difference? Use an appropriate statistical test to explain your answer. In particular, write down the details of the null and alternative hypotheses (with the appropriate regression models), the formula of the test statistic, the critical value, and your conclusion.

Results of Regressions of Average Weekly Earnings on Gender and Education Binary Variables and Other Characteristics Using 2007 Data from a Developing Country Survey

Dependent variable: log average weekly earnings (AWE).

Regressor	(1)	(2)	(3)
High school graduate (X_1)	0.352 (0.021)	0.373 (0.021)	0.371 (0.021)
Male (X_2)	0.458 (0.021)	0.457 (0.020)	0.451 (0.020)
Age (X_3)		0.011 (0.001)	0.011 (0.001)
North (X_4)			0.175 (0.037)
South (X_5)			0.103 (0.033)
East (X_7)			-0.102 (0.043)
Intercept	12.84 (0.018)	12.471 (0.049)	12.390 (0.057)
Summary Statistics and Joint Tests			
F -statistic for regional effects = 0			21.87
SER	1.026	1.023	1.020
R^2	0.0710	0.0761	0.0814
n	10973	10973	10973

Question 3 (a)

- i. For testing $H_0: \beta_1 = 0$ vs. $H_1: \beta_1 \neq 0$,

$$|t| = \left| \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} \right| = \left| \frac{0.352 - 0}{0.021} \right| = 16.7619 > 1.96$$

Therefore, the high school earning difference is statistically significant at the 5% level.

The 95% confidence interval for β_1 is: $[\hat{\beta}_1 - 1.96 \times SE(\hat{\beta}_1), \hat{\beta}_1 + 1.96 \times SE(\hat{\beta}_1)]$
 $= [0.352 - 1.96 \times 0.021, 0.352 + 1.96 \times 0.021] = [0.3108, 0.3932]$

- ii. For testing $H_0: \beta_2 = 0$ vs. $H_1: \beta_2 \neq 0$,

$$|t| = \left| \frac{\hat{\beta}_2 - 0}{SE(\hat{\beta}_2)} \right| = \left| \frac{0.458 - 0}{0.021} \right| = 21.8059 > 1.96$$

Therefore, the male-female earning difference is statistically significant at the 5% level.

The 95% confidence interval for β_2 is: $[\hat{\beta}_2 - 1.96 \times SE(\hat{\beta}_2), \hat{\beta}_2 + 1.96 \times SE(\hat{\beta}_2)]$
 $= [0.458 - 1.96 \times 0.021, 0.458 + 1.96 \times 0.021] = [0.4168, 0.4992]$

Question 3 (b)

- i. For testing $H_0: \beta_3 = 0$ vs. $H_1: \beta_3 \neq 0$,

$$|t| = \left| \frac{\hat{\beta}_3 - 0}{SE(\hat{\beta}_3)} \right| = \left| \frac{0.011 - 0}{0.001} \right| = 11 > 1.96$$

Therefore, the coefficient of age is statistically significant at the 5% level.

The 95% confidence interval for β_3 is: $[\hat{\beta}_3 - 1.96 \times SE(\hat{\beta}_3), \hat{\beta}_3 + 1.96 \times SE(\hat{\beta}_3)]$
 $= [0.011 - 1.96 \times 0.001, 0.011 + 1.96 \times 0.001] = [0.009, 0.013]$ which does not contain zero.
 $\Rightarrow$ age is an important determinant of wage.

- ii. Note that in the question, $\Delta Age = 10$ (the difference in Age, where $\Delta Age = \Delta X_3$). Then, according to the given formula, the 95% confidence interval for $\beta_3 \Delta Age$ is

$$\begin{aligned} & \left[\left(\hat{\beta}_3 - 1.96 \times SE(\hat{\beta}_3) \right) \Delta Age, \left(\hat{\beta}_3 + 1.96 \times SE(\hat{\beta}_3) \right) \Delta Age \right] \\ & = [0.009 \times 10, 0.013 \times 10] = [0.09, 0.13] \end{aligned}$$

Question 3 (c)

For column (3), the regression model can be written as:

$$AWE_i = \beta_0 + \beta_1 HighSchool_i + \beta_2 Male_i + \beta_3 Age_i + \beta_4 North_i + \beta_5 South_i + \beta_6 East_i + u_i$$

where $i = 1, \dots, n$. To test if there is any important regional difference, we can form the following hypotheses:

$$H_0: \beta_4 = \beta_5 = \beta_6 = 0, \text{ vs. } H_1: \beta_j \neq 0 \text{ for at least one } j (j = 4, 5, 6)$$

From the **unrestricted regression** model above, we can obtain R_u^2 . The **restricted regression** model, which imposes H_0 , can be written as:

$$AWE_i = \beta_0 + \beta_1 HighSchool_i + \beta_2 Male_i + \beta_3 Age_i + u_i$$

where $i = 1, \dots, n$. And then we obtain R_r^2 from the **restricted model**. Also, we note that $q = 3$, $k = 6$, and $n = 10973$. Therefore, we can define the homoskedasticity-only F -statistic as

$$F = \frac{(R_u^2 - R_r^2)/3}{(1 - R_u^2)/(10973 - 6 - 1)}$$

According to the table of estimation results in the question, $R_u^2 = 0.0814$, $R_r^2 = 0.0761$, so that the homoskedasticity-only F -statistic is equal to 21.09; the heteroskedasticity-robust F -statistic (F -statistic for regional effects = 0) is equal to 21.87. Given that $q = 3$, the 5% critical value for $F_{3,\infty}$ ($\chi_3^2/3$) is 2.6. We have $F > 2.6$ so that H_0 is rejected at the 5% level (no matter using the homoskedasticity-only or the heteroskedasticity-robust F test). Therefore, we conclude that there exists significant regional difference in wages.

Question 4

An economist in US is interested in the impact of Wisconsin's 2011 "Act 10" legislation on wages (which reduced the power of labour unions). She computes the following statistics for average hourly wage rates in Wisconsin and the neighbouring state of Minnesota for the decades before and after Act 10 was enacted (see the table below). Then, she estimates the effect of the "Act 10" legislation on wages by using the following formula:

$$(16.72 - 15.23) - (18.10 - 16.42) = -0.19$$

- a) Does this estimation formula make sense for you? What would be the key assumption involved in the estimation? Explain.
- b) Give an empirical example of using the "difference-in-differences" (DID) to study the effect of certain policy change or intervention. Show the diagram of the DID design in your example.

	Year	Average Wage
Wisconsin	2001-2010	15.23
Wisconsin	2010-2020	16.72
Minnesota	2001-2010	16.42
Minnesota	2010-2020	18.10

Question 4 (a)

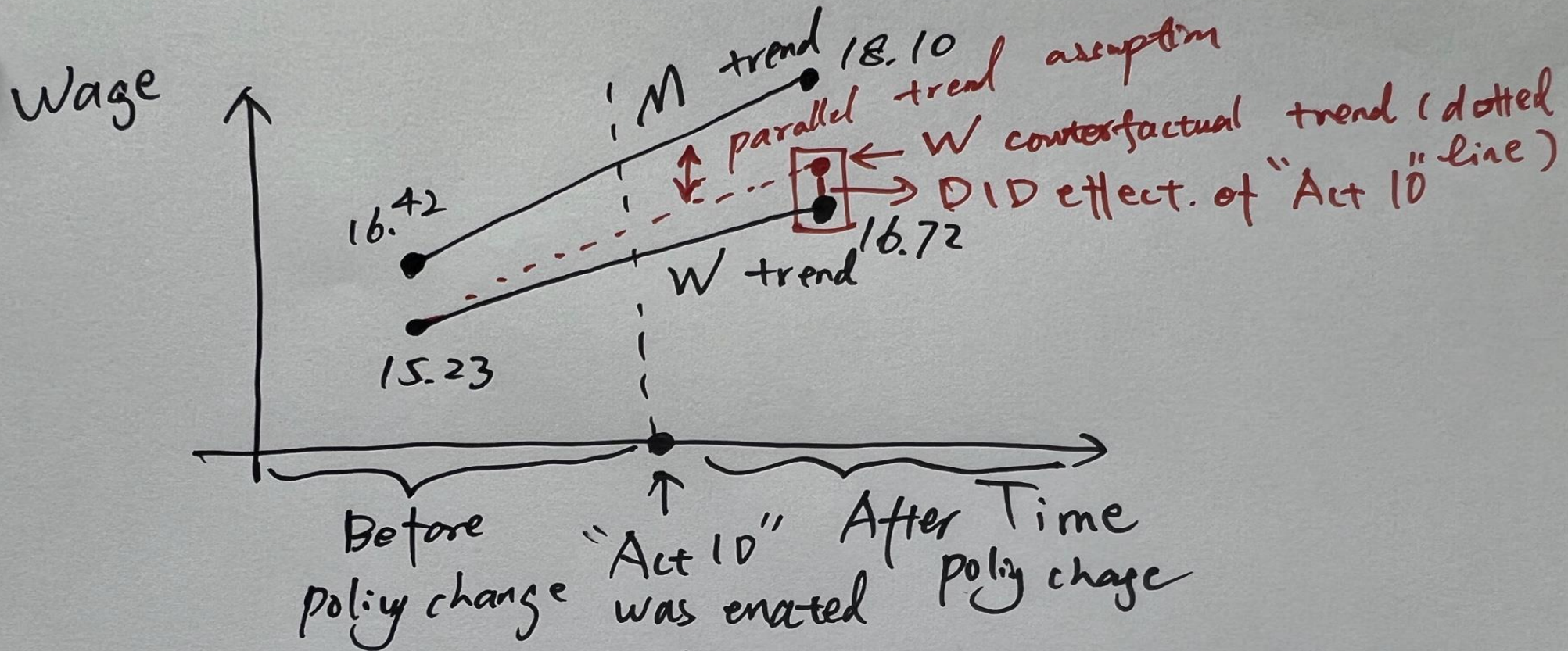
Let avg. denotes average. According to the question, $(16.72 - 15.23) - (18.10 - 16.42)$ has the following meaning (that is , **Wisconsin Trend – Minnesota Trend**):

$$\begin{aligned} & (\text{avg.wage of Wisconsin after the legislation} - \text{avg.wage of Wisconsin before the legislation}) \\ & - (\text{avg.wage of Minnesota after the legislation} - \text{avg.wage of Minnesota before the legislation}) \end{aligned}$$

which can be seen as:

$$\begin{aligned} & (\text{change in avg.wage due to US macroeconomy} + \text{the effect of the legislation}) \\ & - (\text{change in avg.wage due to US macroeconomy}) \\ & = \text{the effect of the legislation} \end{aligned}$$

Therefore, the difference-in-differences (DID) estimate wipes out the effect due to the US macroeconomy so that we can find out what is the net effect of the Act 10 legislation. According to **this equation**, the key assumption is that during this time period, **the US macroeconomy affected the average wage in Wisconsin and Minnesota in the same way (parallel trend assumption)**.



Note: Wisconsin (W) serves as the treatment state while Minnesota (M) serves as the control state.

Question 4 (b)

Study the effect of free lunch program on academic performance (Gordanier et al., Economics of Education Review, 2020).

- Study the effect of the Community Eligibility Provision (CEP), a free-lunch program, on elementary and middle school students' academic performance in South Carolina.
- Treatment group: students from eligible schools that receive free lunches
- Control group: students from schools that do not participate in the program
- Before/After: before/after the introduction of the free lunch program
- Measure student's performance using math and reading test scores

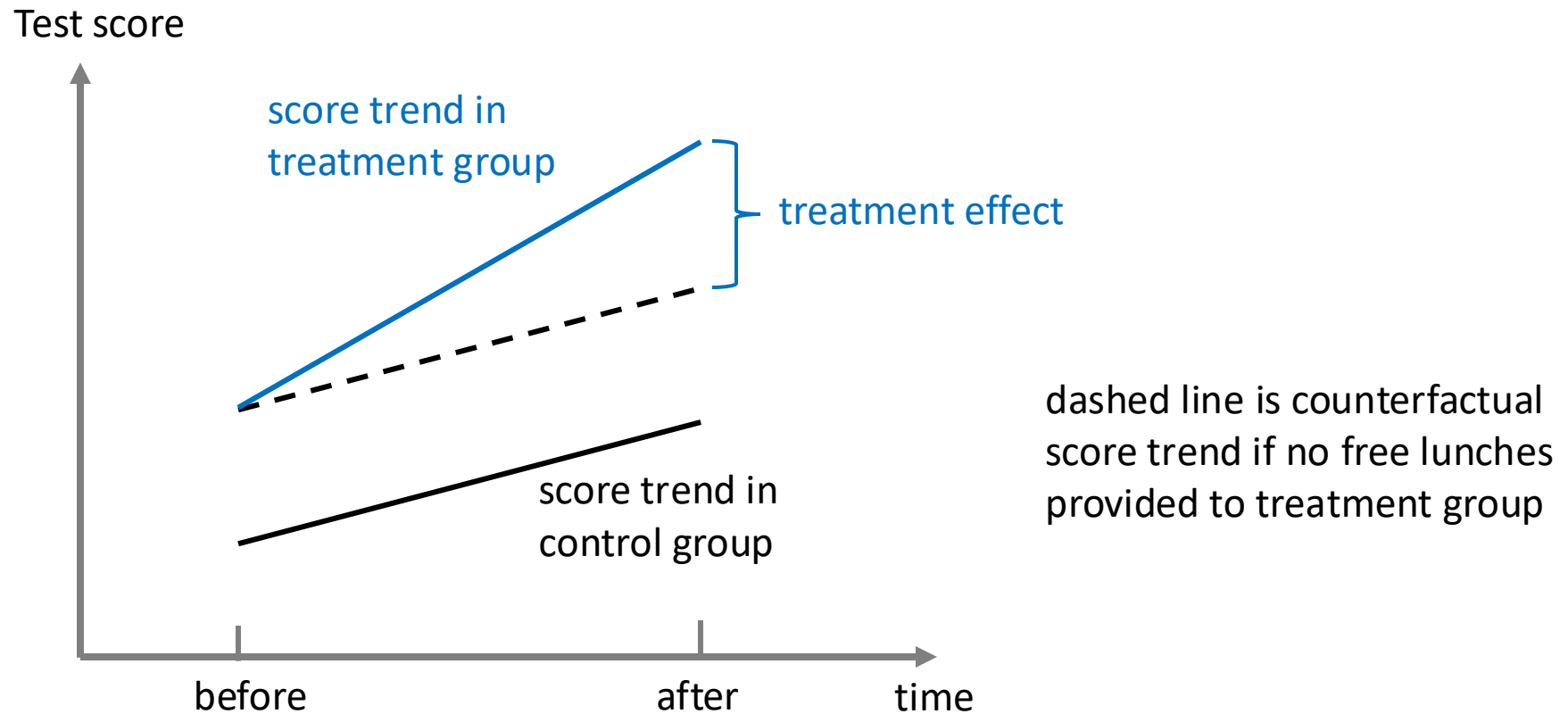
Denote $\bar{Y}$ as the average test score, we have the treatment effect of free lunch program:

$$\text{Treatment Effect} = (\bar{Y}_{treat,after} - \bar{Y}_{treat,before}) - (\bar{Y}_{control,after} - \bar{Y}_{control,before})$$

which can be seen as:

$$\begin{aligned} &= (\text{change in } \bar{Y} \text{ due to learning process} + \text{the effect of free lunch}) \\ &- (\text{change in } \bar{Y} \text{ due to learning process}) \end{aligned}$$

Question 4 (b)



Question 5

Empirical Exercise: Use the dataset **growth.dta** uploaded in NTU Learn, which contains data on average economic growth rate for 65 countries, along with variables that are potentially related to growth. A detailed description of the variables is given in the table below.

- a) Construct a scatterplot of average annual growth rate (*growth*) and average trade share (*tradeshare*). For example, for STATA we can use: **scatter growth tradeshare**. Suppose that we want to run a regression of *growth* on *tradeshare*. According to the scatterplot, one country, Malta, has a *tradeshare* much larger than the other countries. Explain how Malta may affect the regression result.
- b) Run a multiple regression of growth on the following variables: *tradeshare*, *yearsschool*, *rev_coups*, *assassinations*, and *rgdp60*. What is the estimated value of the coefficient on *yearsschool*? Is the coefficient significant at the 5% level? Explain using t-test, p-value and confidence interval.
- c) Why is *oil* omitted from the regression? What would happen if it were included?
- d) Test the joint hypothesis that none of the political/criminal factors has effect on economic *growth* rate. What is your conclusion?

Question 5

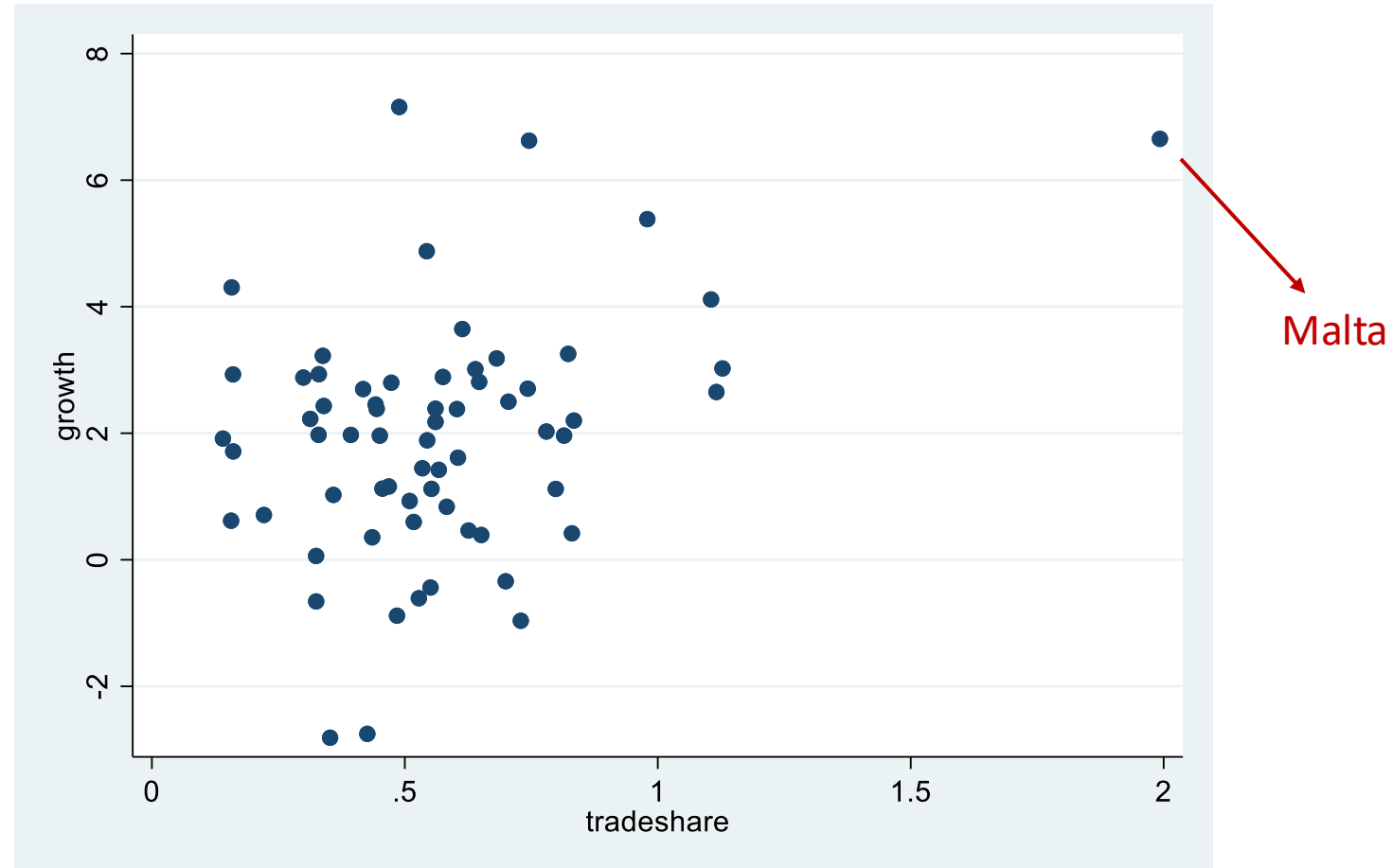
Variable Definitions

Variable	Definition
<i>Country_name</i>	Name of country
<i>growth</i>	Average annual percentage growth of real Gross Domestic Product (GDP)* from 1960 to 1995.
<i>rgdp60</i>	The value of GDP* per capita in 1960, converted to 1960 US dollars
<i>tradehare</i>	The average share of trade in the economy from 1960 to 1995, measured as the sum of exports plus imports, divided by GDP; that is, the average value of $(X + M)/GDP$ from 1960 to 1995, where X = exports and M = imports (both X and M are positive).
<i>yearsshcool</i>	Average number of years of schooling of adult residents in that country in 1960
<i>rev_coups</i>	Average annual number of revolutions, insurrections (successful or not) and coup d'etats in that country from 1960 to 1995
<i>assasinations</i>	Average annual number of political assassinations in that country from 1960 to 1995 (per million population)
<i>oil</i>	= 1 if oil accounted for at least half of exports in 1960 = 0 otherwise

Question 5 (a)

As we can see in the graph there is one outlier Malta, which has much larger value of *tradeshare*. From OLS Assumption 3 we know that if there exists outliers, the OLS estimator will be biased. This can also be seen by comparing the OLS estimates with and without Malta.

Stata command: **scatter growth tradeshare**



Question 5 (a)

```
. reg growth tradeshare, robust
```

Linear regression

Number of obs = 65
F(1, 63) = 12.09
Prob > F = 0.0009
R-squared = 0.1237
Root MSE = 1.79

Full Sample

growth	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
<u>tradeshare</u>	<u>2.306434</u>	.6632868	3.48	0.001	.9809608	3.631907
_cons	.6402653	.4591457	1.39	0.168	-.2772641	1.557795

Question 5 (a)

```
. reg growth tradeshare if country != "Malta", robust
```

Linear regression

Number of obs = 64 Excluding Malta
F(1, 62) = 3.77
Prob > F = 0.0567
R-squared = 0.0447
Root MSE = 1.7894

growth	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
<u>tradeshare</u>	<u>1.680905</u>	.8656171	1.94	0.057	-.0494392	3.411249
_cons	.9574107	.5360579	1.79	0.079	-.1141537	2.028975

Question 5 (b)

```
. reg growth tradeshare yearsschool rev_coups assassinations rgdp60, robust
```

Linear regression	Number of obs	=	65
	F(5, 59)	=	11.20
	Prob > F	=	0.0000
	R-squared	=	0.3589
	Root MSE	=	1.5821

growth	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
tradeshare	1.561696	.584982	2.67	0.010	.3911494	2.732242
yearsschool	.5748461	.1228068	4.68	0.000	.3291101	.820582
rev_coups	-2.157503	.8782778	-2.46	0.017	-3.914933	-.400073
assassinations	.3540784	.3654816	0.97	0.337	-.3772485	1.085405
rgdp60	-.0004693	.0001192	-3.94	0.000	-.000708	-.0002307
_cons	.4897603	.7379584	0.66	0.509	-.9868911	1.966412

Given the coefficient for *yearsschool*:
 $t\text{-stat} = 4.68 > 1.96$, and
 $p\text{-value} = 0.000 < 0.05$,
and the 95% confidence interval does not include 0.
Thus, we can reject H_0 at the 5% level and the coefficient for *yearsschool* is statistically significant.

Question 5 (c)

```
. summarize oil
```

Variable	Obs	Mean	Std. dev.	Min	Max
oil	65	0	0	0	0

oil is omitted by STATA because of multicollinearity. The OLS regression cannot be run if it were included. By summarizing *oil* variable or checking the “Data Editor” in STATA, we can see that *oil* is equal to 0 for all the observations.

	country_name	growth	oil	rgdp60	tradeshare	yearsschool	rev_coups	assasinations
1	India	1.915168	0	765.9998	.140502	1.45	.1333333	.8666667
2	Argentina	.6176451	0	4462.001	.156623	4.99	.9333333	1.933333
3	Japan	4.304759	0	2954	.1577032	6.71	0	.2
4	Brazil	2.930097	0	1784	.1604051	2.89	.1	.1
5	United States	1.712265	0	9895.004	.160815	8.66	0	.4333333
6	Bangladesh	.7082631	0	951.9998	.2214584	.79	.3064815	.175
7	Spain	2.880327	0	3123.001	.2994059	3.8	.0666667	1.433333
8	Colombia	2.227014	0	1684	.3130729	2.97	.1	.7666667
9	Peru	.060206	0	2019	.3246127	3.02	.2666667	.5666667
10	Haiti	-.6579341	0	923.9999	.3247456	.7	.3740741	.2

Question 5 (d)

```
. reg growth tradeshare yearsschool rev_coups assassinations rgdp60, robust
```

Linear regression

Number of obs = 65
 F(5, 59) = 11.20
 Prob > F = 0.0000
 R-squared = 0.3589
 Root MSE = 1.5821

growth	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
tradeshare	1.561696	.584982	2.67	0.010	.3911494	2.732242
yearsschool	.5748461	.1228068	4.68	0.000	.3291101	.820582
rev_coups	-2.157503	.8782778	-2.46	0.017	-3.914933	-.400073
assassinations	.3540784	.3654816	0.97	0.337	-.3772485	1.085405
rgdp60	-.0004693	.0001192	-3.94	0.000	-.000708	-.0002307
_cons	.4897603	.7379584	0.66	0.509	-.9868911	1.966412

Question 5 (d)

After running regression in Stata, we can conduct the F -test. As can be seen below, the p -value of the heteroskedasticity-robust F -test is equal to 0.0546. Therefore, the joint null hypothesis that none of the political/criminal factors has effect on economic growth rate can be rejected at the 10% level but not 5% level.

```
. test rev_coups assassinations
```

```
( 1)  rev_coups = 0
```

```
( 2)  assassinations = 0
```

```
      F(  2,    59) =    3.06  
      Prob > F =    0.0546
```